

Lab5

Introduction to Jenkins: What is Jenkins? Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use.

Introduction to Jenkins

What is Jenkins?

Jenkins is an open-source automation server widely used in the field of Continuous Integration (CI) and Continuous Delivery (CD). It allows developers to automate the building, testing, and deployment of software projects, making the development process more efficient and reliable. It provides a web-based interface and supports a wide range of plugins for integrating with different tools and technologies.

Key features of Jenkins:

- **CI/CD:** Jenkins supports Continuous Integration and Continuous Deployment, allowing developers to integrate code changes frequently and automate the deployment of applications.
- **Plugins:** Jenkins has a vast library of plugins that can extend its capabilities. These plugins integrate Jenkins with version control systems (like Git), build tools (like Maven or Gradle), testing frameworks, deployment tools, and much more.
- **Pipeline as Code:** Jenkins allows the creation of pipelines using Groovy-based DSL scripts or YAML files, enabling version-controlled and repeatable pipelines.
- **Cross-platform:** Jenkins can run on various platforms such as Windows, Linux, macOS, and others.

Installing Jenkins

Jenkins can be installed on local machines, on a cloud environment, or even in containers. Here's how you can install Jenkins in Window local System environments:

1. Installing Jenkins Locally

1. Prerequisites:

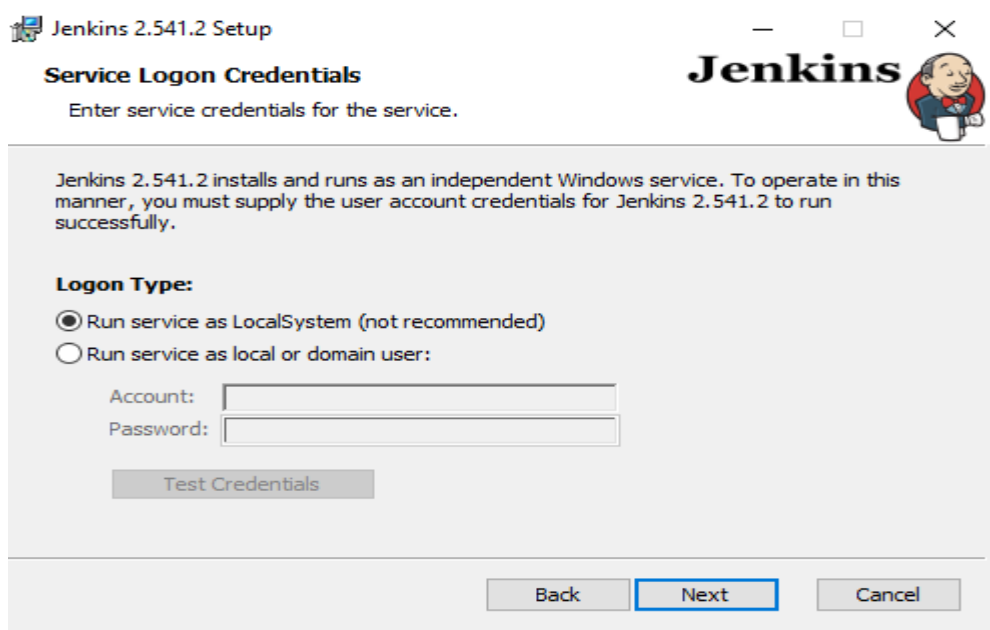
- Ensure that **Java (JDK)** is installed on your system. **Jenkins requires Java 21.** If not then [click here](#).
- You can check if Java is installed by running **java -version** in the terminal.

2. Install Jenkins on Window System):

- Download the Jenkins Windows installer from the [official Jenkins website](#).



- Run the installer and follow the on-screen instructions. While installing choose **login system: run service as LocalSystem (not recommended)**.



- After then use **default port** or you can **configure you own port like I'm using port 3030** then click on test and next.

a) Using Command Prompt (check any process is running on port 3030)

Open **Command Prompt** as administrator and run:

```
netstat -ano
```

```
C:\Users\Student>netstat -ano

Active Connections

  Proto Local Address           Foreign Address         State       PID
  TCP    0.0.0.0:135               0.0.0.0:0               LISTENING   8
  TCP    0.0.0.0:445               0.0.0.0:0               LISTENING   4
  TCP    0.0.0.0:623               0.0.0.0:0               LISTENING   3472
  TCP    0.0.0.0:1521              0.0.0.0:0               LISTENING   3352
  TCP    0.0.0.0:2343              0.0.0.0:0               LISTENING   4140
  TCP    0.0.0.0:3306              0.0.0.0:0               LISTENING   4668
  TCP    0.0.0.0:3580              0.0.0.0:0               LISTENING   3308
  TCP    0.0.0.0:5040              0.0.0.0:0               LISTENING   7140
  TCP    0.0.0.0:7680              0.0.0.0:0               LISTENING   6276
  TCP    0.0.0.0:16992             0.0.0.0:0               LISTENING   3472
  TCP    0.0.0.0:33060             0.0.0.0:0               LISTENING   4668
```

Explanation:

- `netstat` → Network statistics
- `-a` → Shows all connections and listening ports
- `-n` → Shows addresses and port numbers in numeric form
- `-o` → Shows the **Process ID (PID)** using each port

You'll see output like:

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:80	0.0.0.0:0	LISTENING	1234
TCP	127.0.0.1:3306	0.0.0.0:0	LISTENING	5678

Find the process using a port:

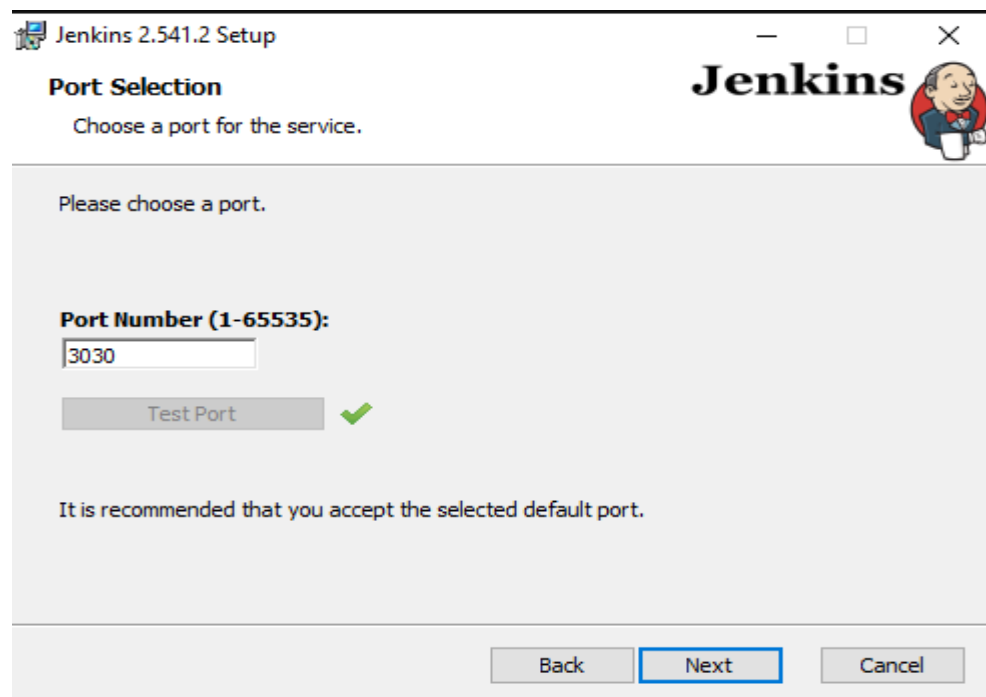
```
tasklist /FI "PID eq 1234"
```

```
C:\Users\Student>tasklist /FI "PID eq 3352"

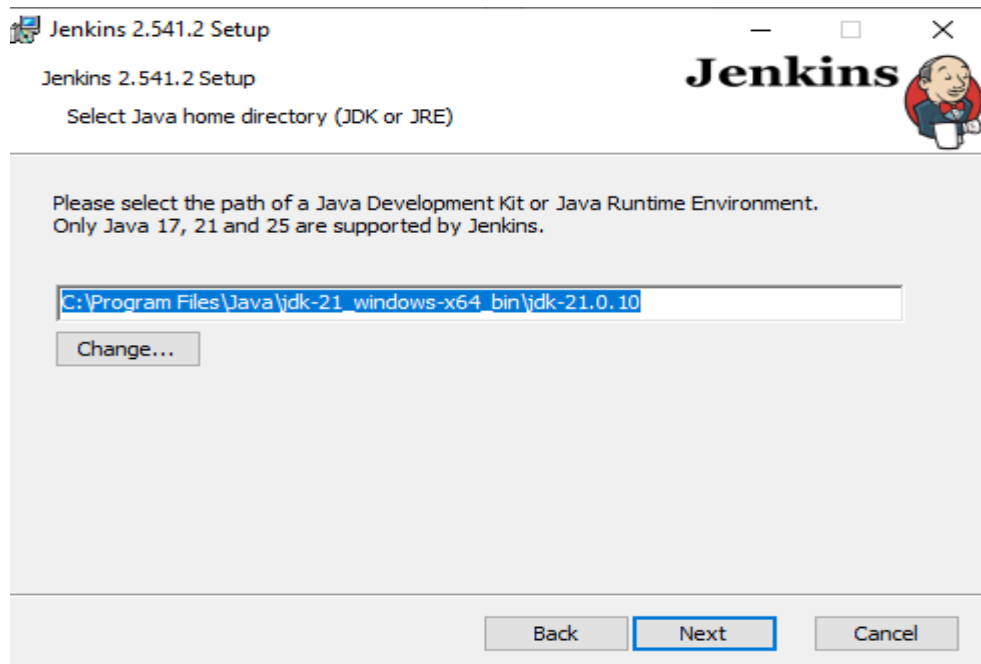
Image Name                        PID Session Name        Session#    Mem Usage
=====
TNSLNR.EXE                        3352 Services              0         10,748 K

C:\Users\Student>tasklist /FI "PID eq 8080"
INFO: No tasks are running which match the specified criteria.

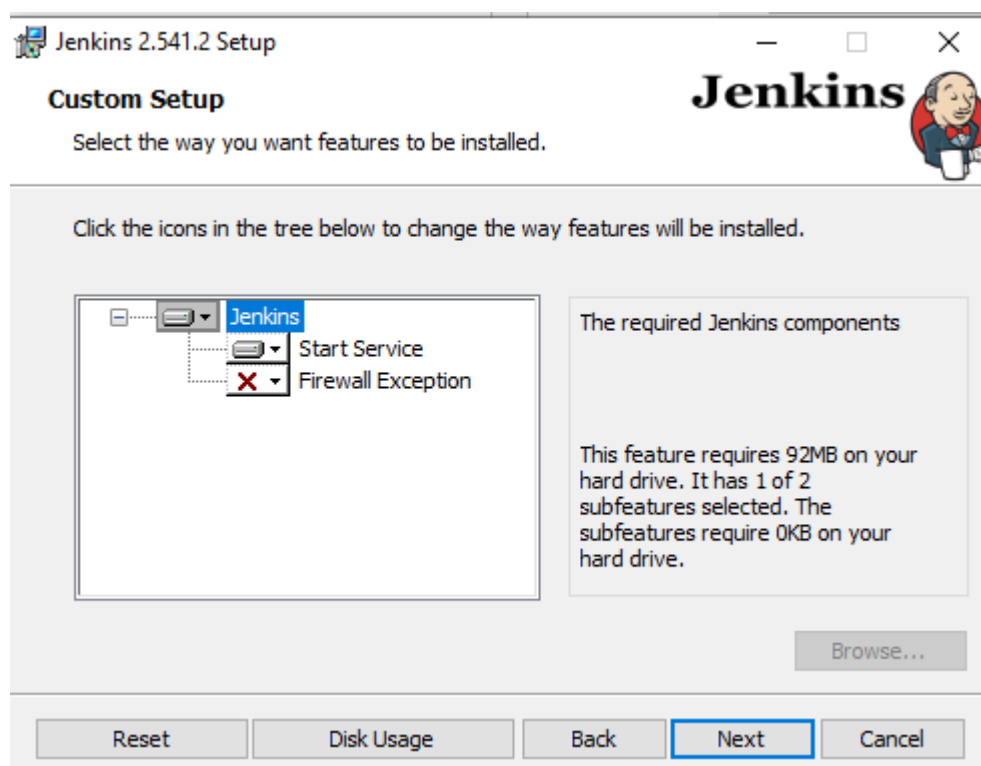
C:\Users\Student>tasklist /FI "PID eq 3030"
INFO: No tasks are running which match the specified criteria.
```

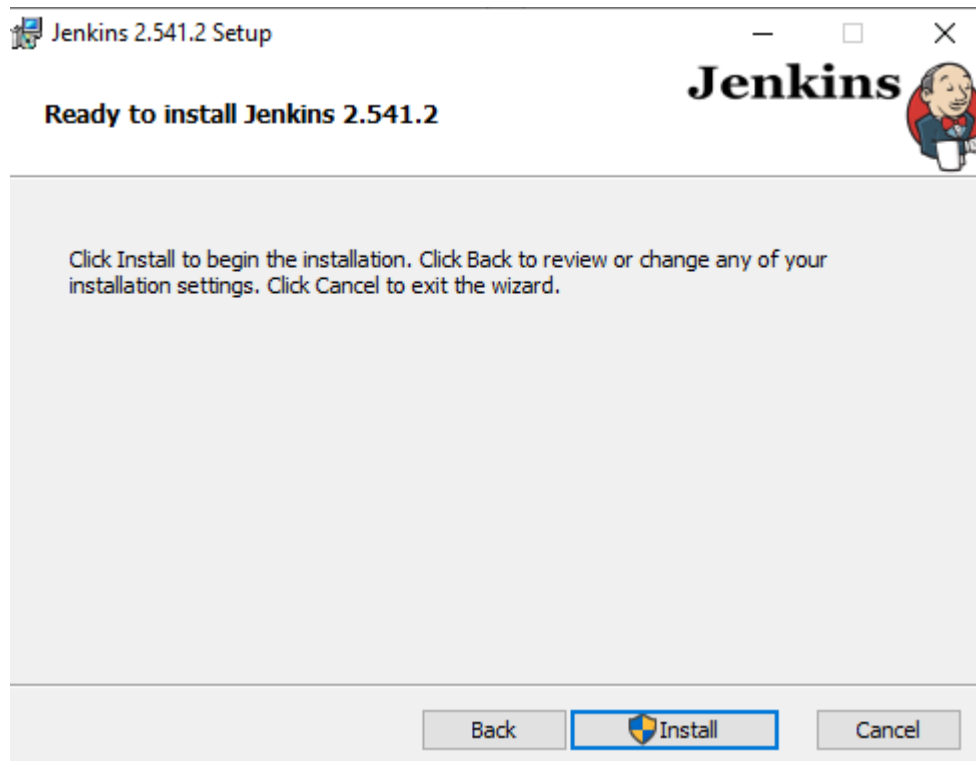


- After then **change the directory** and choose **java jdk-21** path look like **C:\Program Files\Java\jdk-21**.



- After then **click next, next** and then it will **ask permission** click on **yes** and it will start installing.

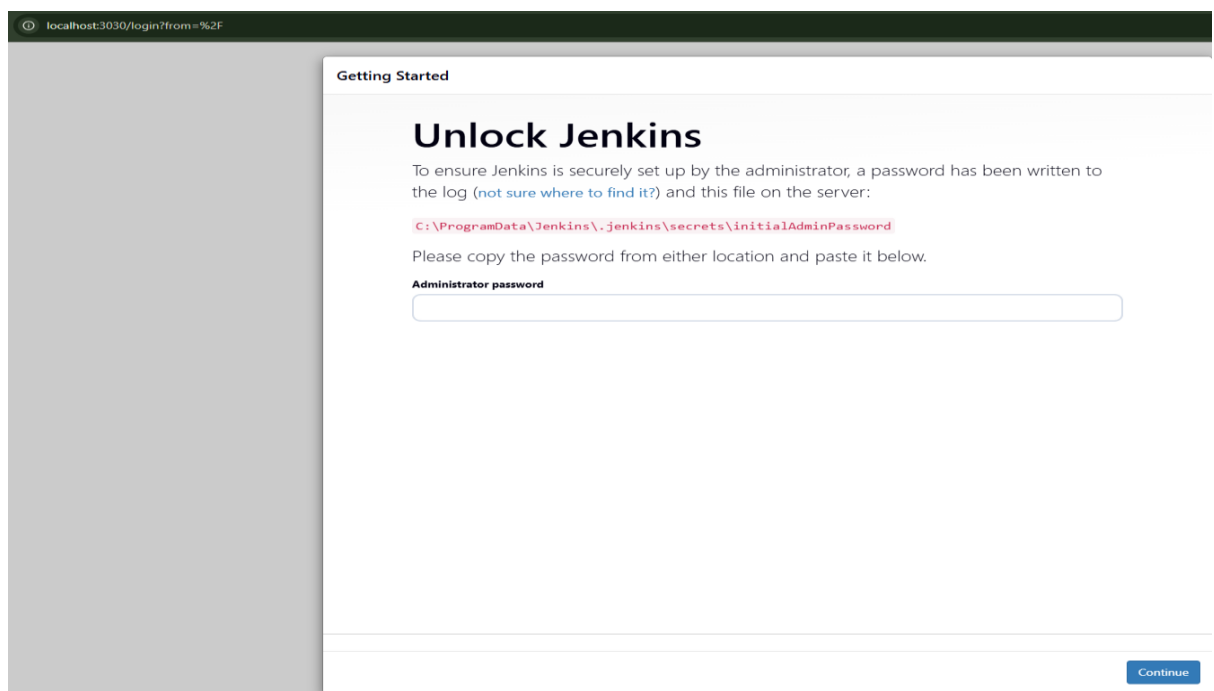




- After successfully installed, Jenkins will be **running on port either default port or chosen port 3030** by default (you can access it in your browser at **http://localhost:8080**) or **http://localhost:3030**.

2. Jenkins Setup in browser:

- After opening browser by visiting your local address the browser should look like below screenshot.



- It will ask **administrator password** so you have to **navigate the above highlighted path** and open that **initialAdminPassword** in **notepad** or any software to **see the password**.

```
C:\Users\Student>type C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
```

- Just **copy that password** and **paste it** and **click on continue**.
- It will ask to **customize Jenkins** so **click on install suggested plugin** it will **automatically install all required plugin**.

Getting Started

Getting Started



☒ Folders	☒ OWASP Markup Formatter	☐ Build Timeout	☐ Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API Folders OWASP Markup Formatter ** ASM API
☐ Timestampers	☐ Workspace Cleanup	☐ Ant	☐ Gradle	
☐ Pipeline	☐ GitHub Branch Source	☐ Pipeline: GitHub Groovy Libraries	☐ Pipeline Graph View	
☐ Git	☐ SSH Build Agents	☐ Matrix Authorization Strategy	☐ LDAP	
☐ Email Extension	☐ Mailer	☐ Dark Theme		

- After then **create admin profile** by **filling all details** then **click on save and continue** after then **save and finish** after then **click on start using Jenkin**.

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

Start using Jenkins

- We can access it in the browser at **`http://localhost:8080`** or **`http://localhost:3030`**.

